HOTEL BOOKING DEMAND DATASET ANALYSIS

Technical Report

Cleaning and Feature Engineering

Executive Summary

I analyzed the Hotel Booking Demand dataset to understand booking behavior, identify data-quality issues, perform exploratory analysis, investigate outliers, clean the data, and create useful derived features. The original dataset contained 119,390 observations and 32 variables covering hotel type, arrival information, stay duration, guest characteristics, booking channels, room information, deposit type, reservation status, price, and special requests.

I first performed data understanding and data-quality investigation. I examined the structure, feature meanings, data types, summary statistics, missing values, duplicate records, inconsistent categories, and columns requiring special attention. The original data contained 31,994 exact duplicate rows.

I then performed univariate, categorical, numerical, bivariate, and correlation analysis. The analysis showed clear cancellation patterns: City Hotel had more cancellation activity, longer lead times were linked with higher cancellation, repeated guests were more reliable, and special requests were associated with lower cancellation.

For outlier analysis, I used both IQR and Z-score methods. I compared their results and investigated extreme values instead of automatically deleting every statistical outlier. Valid extreme observations were preserved.

During cleaning, I treated missing values, duplicates, categories, data types, and unnecessary columns systematically. During final verification, seven duplicated pairs remained, representing 14 rows; these were removed, leaving zero duplicate rows.

Finally, I created four engineered features: total_stay_nights, total_guests, is_family, and is_long_stay. The final dataset contains 87,389 rows and 36 columns, with zero missing values and zero duplicate rows.

1. Dataset and Background

The Hotel Booking Demand dataset contains booking records from City Hotel and Resort Hotel. Each observation includes information about booking timing, arrival, stay duration, guests, meal, country, market segment, distribution channel, room type, deposit type, customer type, price, special requests, and reservation outcome.

I used this dataset to study hotel booking and cancellation behavior and to prepare a reliable dataset for later analytical or predictive work.

Item	Value
Original rows	119,390
Original columns	32
Final rows	87,389
Final columns	36
Engineered features	4
Final missing values	0
Final duplicate rows	0

2. Dataset Overview

The main outcome was is_canceled, where 1 represents a canceled booking and 0 represents a booking that was not canceled. The dataset combines numerical measurements, categorical variables, identifier-like fields, and date information.

3. Data Understanding

3.1 Introduction

I began by understanding the structure and meaning of the dataset before making cleaning decisions. This helped me distinguish dates, identifiers, categories, counts, and continuous measurements.

3.2 Import Libraries

I used pandas and NumPy for data handling, Matplotlib and Seaborn for visualization, SciPy for statistical calculations, and scikit-learn for outlier-related analysis.

3.3 Load Dataset

I loaded hotel_bookings.csv into a pandas DataFrame and used it as the main object for inspection and analysis.

3.4 Dataset Overview

The original shape was (119390, 32). I inspected the first and last rows, column names, and general structure.

3.5 Feature Description

The features describe hotel type, cancellation, lead time, arrival dates, stay length, guest counts, meal, country, market segment, distribution channel, previous booking history, room types, deposit type, customer type, ADR, special requests, and reservation status.

3.6 Data Types

The initial data contained 16 integer, 4 float, and 12 object columns. reservation_status_date was stored as object even though it represented dates. Agent and company were float because of missing values but behaved as identifiers. Children was float because of its four missing observations.

3.7 Summary Statistics

I reviewed count, mean, standard deviation, minimum, quartiles, and maximum for numerical variables and frequency information for categorical variables. This provided an initial view of distributions and unusual values.

4. Data Quality Investigation

4.1 Missing Value Analysis

I calculated the number and percentage of missing values for every feature. Missingness was concentrated in company, agent, country, and children. I treated these fields according to their meaning instead of applying one rule to all missing values.

Feature	Finding
company	Large missing proportion; identifier-like
Agent	Missing values; identifier-like
Country	488 missing values
Children	4 missing values

4.2 Duplicate Detection

I found 31,994 exact duplicate rows, representing 26.80% of the original dataset. I inspected the repeated observations before treatment and considered them redundant exact records.

4.3 Inconsistent Categories

Most categories were consistent. Undefined values were found in meal (1,169), market_segment (2), and distribution_channel (5). Rare room types were retained because rarity alone did not prove an error.

4.4 Data Type Issues

reservation_status_date was an object containing dates and was converted to datetime during cleaning. Agent and company were treated as identifier-like fields, while children was treated as a count variable.

4.5 Unnecessary Columns

I reviewed every column for analytical relevance, redundancy, data quality, and usefulness. Columns were not removed merely because they had missing values.

5. Exploratory Data Analysis (EDA)

5.1 Univariate Analysis

I analyzed numerical variables independently using count plots, histograms, summary statistics, and distribution checks.

5.1.1 Lead Time Distribution

Lead time was strongly right-skewed. Most bookings were made relatively close to arrival, while a smaller number were booked far in advance. The long tail extended beyond 700 days.

5.1.2 Weekend Stay Nights

Most bookings contained zero weekend nights, followed by one- and two-night stays. Longer weekend stays were uncommon and produced a right tail.

5.1.3 Week Stay Nights

Weekday stays were concentrated around short stays, especially one to four nights. Longer stays were less frequent.

5.1.4 Adults per Booking

Two adults were the most common booking size, followed by one adult. Larger groups were less common and some unusually large values were investigated.

5.1.5 Children per Booking

The distribution was strongly concentrated at zero. One or two children were less common, while higher values were rare. Four values were missing.

5.1.6 Babies per Booking

Most bookings had zero babies. One or two babies were uncommon and a few high values were investigated as extreme observations.

5.1.7 Booking Cancellation Status

Both canceled and non-canceled bookings were well represented, making cancellation suitable as the main outcome for comparison.

5.2 Categorical Analysis

I examined hotel type, arrival month, meal type, market segment, distribution channel, deposit type, customer type, reservation status, country, top countries, reserved room type, and assigned room type. City Hotel had more observations. August was among the busiest arrival months. BB was the dominant meal type, Online TA was the largest market segment, TA/TO was the dominant distribution channel, No Deposit was the dominant deposit type, and Transient customers were the largest customer group.

5.3 Numerical Distributions

Summary statistics, skewness analysis, and box plots showed that several count variables were strongly right-skewed. Lead time and stay-related variables had long tails. Box plots helped identify extreme observations for later investigation without automatically treating them as errors.

5.4 Bivariate Analysis

Hotel Type vs Cancellation

City Hotel had more canceled bookings than Resort Hotel, showing different cancellation patterns by hotel type.

Lead Time vs Cancellation

Longer lead times were associated with higher cancellation behavior.

Stay Duration vs Cancellation

Cancellation varied with stay duration, so weekend and weekday nights were considered together.

ADR vs Cancellation

ADR differed between canceled and non-canceled bookings, showing a relationship between price level and cancellation.

Market Segment vs Cancellation

Cancellation proportions varied across market segments, with some segments showing higher cancellation.

Customer Type vs Cancellation

Transient customers accounted for a large share of cancellations and showed a meaningful relationship with cancellation.

Deposit Type vs Cancellation

Deposit type was strongly related to cancellation behavior.

Special Requests vs Cancellation

Bookings with more special requests generally showed lower cancellation.

Repeated Guest vs Cancellation

Repeated guests were generally more reliable and had lower cancellation.

Reserved vs Assigned Room Type

Most bookings were assigned the room type they reserved, although some reassignment occurred.

6. Correlation Analysis

6.1 Lead Time and Cancellation

Lead time had a positive relationship with cancellation, supporting the EDA result.

6.2 Previous Cancellations and Current Cancellation

Previous cancellations were positively associated with current cancellation behavior.

6.3 Total Special Requests and Cancellation

Special requests showed an inverse relationship with cancellation.

6.4 Previous Bookings Not Canceled and Cancellation

A history of bookings that were not canceled was associated with more reliable behavior.

6.5 Stay Duration Variables

Weekend nights and weekday nights were positively related to total stay duration and should be interpreted as related components.

7. Business Questions

- ### Which hotel type has more cancellation?

City Hotel had more cancellation activity.

- ### Does longer lead time increase cancellation?

Yes. Longer lead times were linked with higher cancellation behavior.

- ### Which customer type cancels the most?

Transient customers had the largest cancellation volume.

- ### Which market segment has the highest cancellation?

Cancellation proportions differed across market segments, with some segments showing clearly higher cancellation.

- ### Does deposit type affect cancellation?

Yes. Deposit arrangements showed substantially different cancellation patterns.

- ### Do new guests cancel more than repeated guests**?**

Yes. New guests were generally more likely to cancel than repeated guests.

- ### Does changing the room type affect bookings?

Room reassignment occurred for a smaller group, while most bookings kept the original room type.

- ### Do cancelled bookings have longer lead times?

Yes. Cancelled bookings generally had longer lead times.

8. Key Insights

- City Hotels had more cancellation activity.
- Longer lead time was linked with more cancellation.
- New guests canceled more often than repeated guests.
- Previous cancellation history was an important signal.
- Assigned room type was strongly related to reserved room type.
- Repeated guests were more reliable.
- Special requests were related to lower cancellation.
- Cancelled bookings generally had longer lead times.
- Most bookings kept their reserved room type.

9. Outlier Detection

9.1 Outlier Detection — IQR

I used the Interquartile Range method to identify observations far from the middle 50% of a numerical variable.

Formula: IQR = Q3 − Q1
Lower Bound = Q1 − 1.5 × IQR
Upper Bound = Q3 + 1.5 × IQR

Component	Meaning
Q1	25th percentile
Q3	75th percentile
IQR	Q3 − Q1
Lower bound	Q1 − 1.5 × IQR
Upper bound	Q3 + 1.5 × IQR

Workflow: numerical variable → Q1/Q3 → IQR → lower and upper bounds → count outliers → investigate.

9.2 Outlier Detection — Z-score

I also used Z-score to measure how far an observation was from the mean in standard-deviation units.

Formula: $Z = (x - \mu) / \sigma$

Symbol	Meaning
x	Observed value
μ	Mean
σ	Standard deviation
Z	Standard deviations from mean

Workflow: numerical variable → mean and standard deviation → Z-score → threshold → count potential outliers → investigate.

9.3 Comparison of IQR and Z-score

I compared the results because IQR is based on quartiles while Z-score depends on the mean and standard deviation. IQR is less affected by skewed distributions, whereas Z-score can be influenced by extreme values. Therefore, the methods did not necessarily identify the same observations. I reviewed the mean and standard deviation of the numerical variables to understand these differences.

9.4 Investigate Outliers

I investigated extreme observations in context rather than deleting them automatically. A long lead time can be a legitimate advance booking, and a high ADR can represent a genuine expensive booking.

9.5 Extreme Values Investigation

ADR

ADR was investigated because it is continuous and skewed. High values were treated as candidates for investigation, not automatic errors.

Lead Time

Lead time had a long right tail because some bookings were made far in advance. These values can be legitimate and were preserved.

9.6 Outlier Treatment Decision

- Observations were not removed solely because they were classified as outliers.
- Discrete and zero-inflated variables were retained.
- Legitimate extreme values were preserved.
- Potentially erroneous values were investigated separately.

- If treatment is required for future machine learning, capping or winsorization may be considered for selected continuous variables.
- The clean dataset therefore retains original valid observations.

10. Cleaning

10.1 Data Cleaning

After data understanding, quality investigation, EDA, and outlier investigation, I performed the cleaning operations to create a consistent analytical dataset.

10.2 Missing Value Treatment

Missing values were treated according to variable meaning. The four missing children values were handled during guest-count cleaning. Missing country values were treated as missing/unknown information. Agent and company were handled according to their identifier-like role.

10.3 Duplicate Treatment

The original dataset contained 31,994 duplicate rows. I removed redundant exact duplicate records and checked the dataset again to verify the result.

10.4 Category Standardization

I reviewed the Undefined values in meal, market_segment, and distribution_channel and standardized category treatment according to their meaning and consistency.

10.5 Data Type Correction

reservation_status_date was converted from object to datetime so it could be correctly used as a date field.

10.6 Unnecessary Column Removal

Columns that were redundant or not required for the final analytical dataset were removed according to analytical relevance.

10.7 Final Data Quality Check

I rechecked missing values, duplicate rows, data types, and dataset shape. During final verification, seven duplicated pairs remained, representing 14 rows. I inspected these repeated pairs and removed the 14 redundant rows. The final duplicate check returned zero.

10.8 Cleaning Summary

Check	Result
Original rows	119,390

Original duplicate rows	31,994
Final rows	87,389
Final columns	36
Final missing values	0
Final duplicate rows	0
Date correction	reservation_status_date → datetime

11. Feature Engineering

After cleaning, I created four features from existing variables to make booking information easier to interpret and use.

11.1 Feature 1 — Total Stay Nights

total_stay_nights = stays_in_weekend_nights + stays_in_week_nights. This gives the complete number of nights in a booking.

11.2 Feature 2 — Total Guests

total_guests = adults + children + babies. This summarizes the booking party size.

11.3 Feature 3 — Family Booking

is_family = 1 when children > 0 or babies > 0; otherwise 0. This identifies bookings containing children or babies.

11.4 Feature 4 — Long Stay

is_long_stay identifies bookings meeting the selected long-stay condition based on total_stay_nights.

11.5 Feature Validation

I checked total_stay_nights against weekend plus weekday nights and total_guests against adults plus children plus babies. I also checked is_family against the presence of children or babies and checked is_long_stay against total_stay_nights and the selected threshold.

Feature	Description	Validation
total_stay_nights	Weekend + weekday nights	Matched source variables
total_guests	Adults + children + babies	Matched guest counts
is_family	Children or babies present	Matched family rule
is_long_stay	Long-stay indicator	Matched stay-duration rule

11.6 Feature Engineering Summary

The four features summarize total stay duration, party size, family presence, and long-stay behavior. They make information spread across several columns easier to analyze.

12. Final Dataset and Overall Analysis Summary

The final dataset is the cleaned and feature-engineered version of the Hotel Booking Demand data. The workflow moved from data understanding to quality investigation, EDA, outlier analysis, cleaning, and feature engineering.

Final Measure	Result
Rows	87,389
Columns	36
Missing values	0
Duplicate rows	0
Engineered features	4
reservation_status_date	datetime

12.1 Overall Findings

- Exact duplication was substantial in the original data and required duplicate treatment.
- Missing values were concentrated in company, agent, country, and children.
- reservation_status_date required conversion to datetime.
- Several numerical variables were strongly right-skewed.
- City Hotel showed more cancellation activity.
- Longer lead times were associated with more cancellation.
- Repeated guests were more reliable than new guests.
- Previous cancellation history was an important cancellation signal.
- Special requests were associated with lower cancellation.
- Most bookings kept their reserved room type.
- IQR and Z-score produced different outlier classifications, so outliers were investigated rather than automatically deleted.
- Valid extreme values were retained.
- Four engineered features were added.

12.2 Final Validation

The final validation confirmed 87,389 rows, 36 columns, zero missing values, and zero exact duplicate rows. The final dataset was ready for export and later analytical or machine-learning use.

13. Conclusion

I completed a full data-understanding, data-quality, EDA, outlier, cleaning, and feature-engineering workflow. I combined statistical checks with business interpretation instead of relying only on automatic rules. The final dataset is cleaner, more consistent, and more informative while preserving valid booking behavior and adding four useful analytical features.